

LAPORAN PRAKTIKUM
PEMROGRAMAN BERORIENTASI OBJEK 3



DISUSUN OLEH :
FATHUR RAMADHAN FAHMI
2311533012

PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNOLOGI INFORMASI
UNIVERSITAS ANDALAS
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MEMBUAT FUNGSI CRUD SERVICE DAN COSTUMER

1.Membuat table service di database MySQL seperti ini :

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
☐	1	id	int(11)			No	None		AUTO_INCREMENT
☐	2	jenis	varchar(255)	utf8mb4_general_ci		No	None		
☐	3	harga	double			No	None		
☐	4	status	varchar(50)	utf8mb4_general_ci		No	None		

2.Membuat class ServiceRepo dan codenya seperti ini :

```
1 package DAO;
2
3 import java.sql.Connection;
4
15 public class ServiceRepo implements ServiceDAO {
16     private Connection connection;
17
18
19     final String insert = "INSERT INTO service(id, jenis, harga, status) VALUES(?,?,?,?);";
20     final String select = "SELECT * FROM service;";
21     final String delete = "DELETE FROM service WHERE id=?;";
22     final String update = "UPDATE service SET jenis=?, harga=?, status=? WHERE id=?;";
23
24     public ServiceRepo() {
25         connection = Database.koneksi();
26     }
27
28     @Override
29     public void save(Service service) {
30         PreparedStatement st = null;
31         try {
32             st = connection.prepareStatement(insert);
33             st.setString(1, service.getId());
34             st.setString(2, service.getJenis());
35             st.setString(3, service.getHarga());
36             st.setString(4, service.getStatus());
37             st.executeUpdate();
38         } catch (SQLException e) {
39             e.printStackTrace();
40         } finally {
41             try {
42                 if (st != null) st.close();
43             } catch (SQLException e) {
44                 e.printStackTrace();
45             }
46         }
47     }
48
49     @Override
50     public List<Service> show() {
51         List<Service> services = new ArrayList<>();
52         Statement st = null;
53         ResultSet rs = null;
54         try {
55             st = connection.createStatement();
56             rs = st.executeQuery(select);
57             while (rs.next()) {
58                 Service service = new Service();
59                 service.setId(rs.getString("id"));
60                 service.setJenis(rs.getString("jenis"));
61                 service.setHarga(rs.getString("harga"));
62                 service.setStatus(rs.getString("status"));
63                 services.add(service);
64             }
65         } catch (SQLException e) {
66             Logger.getLogger(ServiceDAO.class.getName()).log(Level.SEVERE, null, e);
67         } finally {
68             try {
69                 if (rs != null) rs.close();
70                 if (st != null) st.close();
71             } catch (SQLException e) {
72                 e.printStackTrace();
73             }
74         }
75         return services;
76     }
77
78     @Override
79     public void update(Service service) {
80         PreparedStatement st = null;
81         try {
82             st = connection.prepareStatement(update);
83             st.setString(1, service.getJenis());
84             st.setString(2, service.getHarga());
85             st.setString(3, service.getStatus());
86             st.setString(4, service.getId());
```

```

78  @Override
79  public void update(Service service) {
80      PreparedStatement st = null;
81      try {
82          st = connection.prepareStatement(update);
83          st.setString(1, service.getJenis());
84          st.setString(2, service.getHarga());
85          st.setString(3, service.getStatus());
86          st.setString(4, service.getId());
87          st.executeUpdate();
88      } catch (SQLException e) {
89          e.printStackTrace();
90      } finally {
91          try {
92              if (st != null) st.close();
93          } catch (SQLException e) {
94              e.printStackTrace();
95          }
96      }
97  }
98
99  @Override
100 public void delete(String id) {
101     PreparedStatement st = null;
102     try {
103         st = connection.prepareStatement(delete);
104         st.setString(1, id);
105         st.executeUpdate();
106     } catch (SQLException e) {
107         e.printStackTrace();
108     } finally {
109         try {
110             if (st != null) st.close();
111         } catch (SQLException e) {
112             e.printStackTrace();
113         }
114     }
115 }
116 }

```

3. Membuat class ServiceDAO dan code seperti ini :

```

1 package DAO;
2
3 import java.util.List;
4
5
6 public interface ServiceDAO {
7     void save(Service service);
8     List<Service> show();
9     void delete(String id);
10    void update(Service service);
11 }
12

```

4. Membuat JFrame untuk Service seperti ini :

The screenshot shows a Java Swing window titled "Service" with a light gray background. It contains three text input fields labeled "Jenis", "Harga", and "Status". Below these fields are four buttons: "Save", "Update", "Delete", and "Cancel". At the bottom of the window is a table with one row of data.

1	cuci	10000.0	menunggu
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5.Membuat method reset pada JFrame seperti ini :

```
public void reset() {  
    txtJenis.setText("");  
    txtHarga.setText("");  
    txtStatus.setText("");  
}
```

6.Membuat instance pada JFrame ServiceFrame

```
ServiceRepo srvc = new ServiceRepo();  
List<Service> ls;  
public String id;
```

7.Membuat fungsi CREATE pada ServiceFrame

```
JButton btnSave = new JButton("Save");  
btnSave.addActionListener(new ActionListener() {  
    public void actionPerformed(ActionEvent e) {  
        Service service = new Service();  
        service.setJenis(txtJenis.getText());  
        service.setHarga(txtHarga.getText());  
        service.setStatus(txtStatus.getText());  
        srvc.save(service);  
        reset();  
        loadTable();  
    }  
});  
btnSave.setBounds(54, 160, 89, 23);  
contentPane.add(btnSave);
```

8.Membuat fungsi READ pada ServiceFrame

```
public void loadTable() {  
    ls = srvc.show();  
    TableService tu = new TableService(ls);  
    tableService.setModel(tu);  
    tableService.getTableHeader().setVisible(true);  
}
```

9. Membuat fungsi UPDATE pada ServiceFrame

```
tableService = new JTable();
tableService.addMouseListener(new MouseAdapter() {
    @Override
    public void mouseClicked(MouseEvent e) {
        id = tableService.getValueAt(tableService.getSelectedRow(), 0).toString();
        txtJenis.setText(tableService.getValueAt(tableService.getSelectedRow(), 1).toString());
        txtHarga.setText(tableService.getValueAt(tableService.getSelectedRow(), 2).toString());
        txtStatus.setText(tableService.getValueAt(tableService.getSelectedRow(), 3).toString());
    }
});
tableService.setBounds(10, 194, 688, 181);
contentPane.add(tableService);
```

Untuk mengupdate table atau mengubah inputan, ketik code dibawah :

```
JButton btnUpdate = new JButton("Update");
btnUpdate.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        Service service = new Service();
        service.setJenis(txtJenis.getText());
        service.setHarga(txtHarga.getText());
        service.setStatus(txtStatus.getText());
        srvc.update(service);
        reset();
        loadTable();
    }
});
btnUpdate.setBounds(153, 160, 89, 23);
contentPane.add(btnUpdate);
```

10. Membuat fungsi DELETE pada ServiceFrame :

```
JButton btnDelete = new JButton("Delete");
btnDelete.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        if(id != null) {
            srvc.delete(id);
            reset();
            loadTable();
        } else {
            JOptionPane.showMessageDialog(null, "Silahkan Pilih Data yang Akan di Hapus");
        }
    }
});
btnDelete.setBounds(375, 160, 89, 23);
contentPane.add(btnDelete);
```

11.Membuat table Costumer di database MySQL seperti ini :

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int(11)		No	None		AUTO_INCREMENT	Change Drop More
☐	2	nama	varchar(128)	utf8mb4_general_ci	No	None			Change Drop More
☐	3	alamat	text	utf8mb4_general_ci	No	None			Change Drop More
☐	4	nohp	varchar(15)	utf8mb4_general_ci	No	None			Change Drop More

12.Membuat class CostumerRepo dan codenya seperti ini :

```
package DAO;

import java.sql.Connection;

public class CustomerRepo implements CustomerDAO {

    private Connection connection;

    final String insert = "INSERT INTO costumer(nama, alamat, nohp) VALUES(?, ?, ?);";
    final String select = "SELECT * FROM costumer;";
    final String update = "UPDATE costumer SET nama=?, alamat=?, nohp=? WHERE id=?;";
    final String delete = "DELETE FROM costumer WHERE id=?;";

    public CustomerRepo() {
        connection = Database.koneksi();
    }

    @Override
    public void save(Costumer costumer) {
        PreparedStatement st = null;
        try {
            st = connection.prepareStatement(insert);
            st.setString(1, costumer.getNama());
            st.setString(2, costumer.getAlamat());
            st.setString(3, costumer.getNohp());
            st.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            try {
                if (st != null) {
                    st.close();
                }
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

```

@Override
public List<Costumer> show() {
    List<Costumer> ls = new ArrayList<>();
    try {
        Statement st = connection.createStatement();
        ResultSet rs = st.executeQuery(select);
        while (rs.next()) {
            Costumer costumer = new Costumer();
            costumer.setId(rs.getString("id"));
            costumer.setNama(rs.getString("nama"));
            costumer.setAlamat(rs.getString("alamat"));
            costumer.setNohp(rs.getString("nohp"));
            ls.add(costumer);
        }
    } catch (SQLException e) {
        Logger.getLogger(CustomerDAO.class.getName()).log(Level.SEVERE, null, e);
    }
    return ls;
}

@Override
public void update(Costumer costumer) {
    PreparedStatement st = null;
    try {
        st = connection.prepareStatement(update);
        st.setString(1, costumer.getNama());
        st.setString(2, costumer.getAlamat());
        st.setString(3, costumer.getNohp());
        st.setString(4, costumer.getId());
        st.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        try {
            if (st != null) {
                st.close();
            }
        } catch (SQLException e) {

```

```

        st.setString(2, costumer.getAlamat());
        st.setString(3, costumer.getNohp());
        st.setString(4, costumer.getId());
        st.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        try {
            if (st != null) {
                st.close();
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}

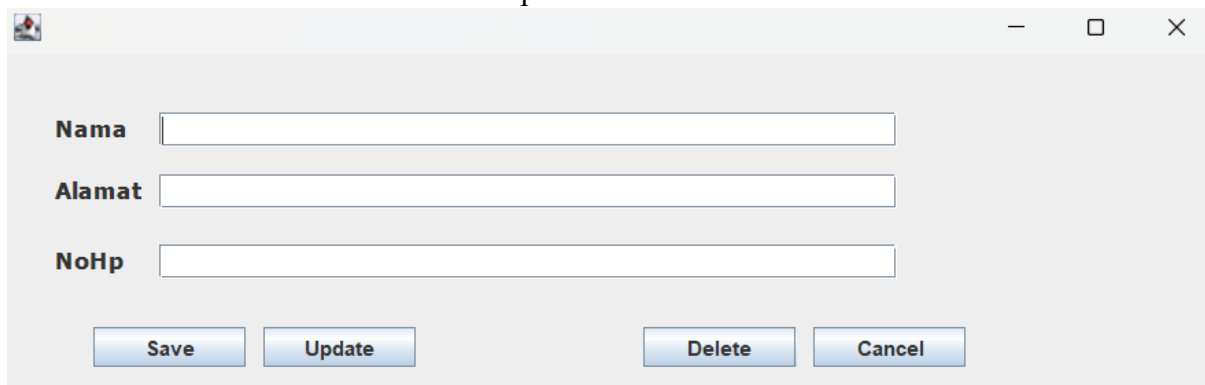
@Override
public void delete(String id) {
    PreparedStatement st = null;
    try {
        st = connection.prepareStatement(delete);
        st.setString(1, id);
        st.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        try {
            if (st != null) {
                st.close();
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
}
}

```


13. Membuat class CostumerDAO dan code seperti ini :

```
1 package DAO;
2
3 import java.util.List;
4
5
6 public interface CustomerDAO {
7     void save(Costumer costumer);
8     List<Costumer> show();
9     void update(Costumer costumer);
10    void delete(String id);
11 }
12
```

14. Membuat JFrame untuk Costumer seperti ini :



The screenshot shows a standard Java Swing window with a title bar containing a maximize button, a close button, and a minimize button. The window has a light gray background. It contains three text input fields stacked vertically, each with a label to its left: 'Nama', 'Alamat', and 'NoHp'. Below these fields, there are four buttons arranged in two groups. The first group contains 'Save' and 'Update' buttons, and the second group contains 'Delete' and 'Cancel' buttons. All buttons have a standard Windows-style appearance with a blue gradient and white text.

15. Membuat method reset pada JFrame seperti ini :

```
public void reset() {
    txtNama.setText("");
    txtAlamat.setText("");
    txtNohp.setText("");
}
```

16. Membuat instance pada JFrame ServiceFrame

```
CustomerRepo cstmr = new CustomerRepo();
List<Costumer> ls;
public String id;
```

17.Membuat fungsi CREATE pada ServiceFrame

```
JButton btnSave = new JButton("Save");
btnSave.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        Costumer costumer = new Costumer();
        costumer.setNama(txtNama.getText());
        costumer.setAlamat(txtAlamat.getText());
        costumer.setNohp(txtNohp.getText());
        cstmr.save(costumer);
        reset();
        loadTable();
    }
});
btnSave.setBounds(54, 160, 89, 23);
contentPane.add(btnSave);
```

18.Membuat fungsi READ pada ServiceFrame

```
public void loadTable() {
    ls = cstmr.show();
    TableCustomer tu = new TableCustomer(ls);
    tableCostumer.setModel(tu);
    tableCostumer.getTableHeader().setVisible(true);
}
}
```

19.Membuat fungsi UPDATE pada ServiceFrame

```
tableCostumer = new JTable();
tableCostumer.addMouseListener(new MouseAdapter() {
    @Override
    public void mouseClicked(MouseEvent e) {
        id = tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 0).toString();
        txtNama.setText(tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 1).toString());
        txtAlamat.setText(tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 2).toString());
        txtNohp.setText(tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 3).toString());
    }
});
tableCostumer.setBounds(10, 194, 688, 228);
contentPane.add(tableCostumer);
```

```

JButton btnUpdate = new JButton("Update");
btnUpdate.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        Costumer costumer = new Costumer();
        costumer.setNama(txtNama.getText());
        costumer.setAlamat(txtAlamat.getText());
        costumer.setNohp(txtNohp.getText());
        cstmr.update(costumer);
        reset();
        loadTable();
    }
});
btnUpdate.setBounds(153, 160, 89, 23);
contentPane.add(btnUpdate);

```

20. Membuat fungsi DELETE pada ServiceFrame :

```

JButton btnDelete = new JButton("Delete");
btnDelete.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        if(id != null) {
            cstmr.delete(id);
            reset();
            loadTable();
        } else {
            JOptionPane.showMessageDialog(null, "Silahkan Pilih Data yang Akan di Hapus");
        }
    }
});
btnDelete.setBounds(375, 160, 89, 23);
contentPane.add(btnDelete);

```

21. Tampilan pada DATABASE :

	id	jenis	harga	status
☐ Edit ☐ Copy ☐ Delete	1	cuci	10000	menunggu

	id	nama	alamat	nohp
☐ Edit ☐ Copy ☐ Delete	1	irfan	padang	02850923852